

RAHUL GUPTA

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EDUCATION

University of Texas at Austin

Expected Graduation: Spring 2028

Bachelor of Science in Aerospace Engineering

GPA: 3.68

- Applied Thermodynamics, Statics, Mechanics of Solids, Low-Speed Aerodynamics, Structural Analysis, Linear Systems, Incompressible Flow, Spacecraft Dynamics*, Propulsion*, Materials Lab*, Flight Dynamics* (*Fall 2026)
- Jefferson Scholar Program

EXPERIENCE

Saronic Technologies | Austin, TX

Associate Test Engineer

Fall 2026 – Present

- Lead continued development and testing of an NDA-protected project initiated during the summer internship, spanning HITL validation, vehicle integration, and vehicle-level testing
- Diagnose and validate vehicle subsystems using power supplies, load banks, multimeters, thermocouples, micrometers, thermal imaging, and CAN bus readers to isolate electrical faults and verify test data

R&D and Test Engineering Intern

Summer 2026 – Fall 2026

- Tested and integrated vehicle upgrades across HITL and vehicle environments, working with vehicle harnessing, CAN bus telemetry, battery and voltage systems, and fluid and tubing assemblies
- Prototyped mounts through additive manufacturing and transitioned final designs to sheet-metal fabrication for production

Longhorn Racing FSAE Electric | UT Austin

Aerodynamics Analysis Lead

Summer 2026 – Present

- Lead four component owners through design reviews, schedule milestones, and cross-component integration
- Coordinate with Dynamics, VMOD, Powertrain, and Body teams to develop aerodynamic targets through trade-off studies
- Develop Python-integrated ANSYS Fluent batch scripts to map full-car aerodynamic performance across the vehicle performance envelope: varying pitch, roll, yaw, ride height, and steering angle
- Lead wind-tunnel and on-track validation, integrating PPS AeroStrip and Telemetry/DAQ measurements to correlate CFD predictions and aeromaps with experimental data

Aerodynamics Analysis Member

Fall 2024 – Spring 2026

- Led rear-wing development from concept through manufacture, applying literature, CFD, wind-tunnel testing, on-track testing, FEA, and cross-system trade-off studies to validate design decisions
- Led the team's first wind-tunnel correlation campaign; secured a \$7,500 PPS partnership and used Python to convert pitot-rake and surface-pressure data into aerodynamic forces and coefficients
- Optimized undertray performance in CFD by mapping ride height variations, increasing simulated vehicle downforce by 10%

AIAA Design Competition | UT Austin

Vehicle Team & Aerodynamics Lead

Fall 2024 – Summer 2025

- Spearheaded aerodynamic design and authored technical analysis sections of a 65-page proposal, optimizing design parameters for a next-generation supersonic interceptor
- Directed the team through critical design review with faculty and senior mentors, incorporating feedback to strengthen the team's final competition submission

PROJECTS

Author – Efficient Methods for Improving Space Science Education

Fall 2023 – Spring 2024

- Designed and executed an IRB-approved research study on aerospace education methodologies, culminating in publication in Exploratio Academic Journal

Author – Denver to Anchorage: A Detailed Optimization of Flight Variables

Summer 2023 – Fall 2023

- Published an analytical paper detailing optimal flight path between major airports using MATLAB computational modeling and fixed-wing aerodynamic principles

SKILLS & ADDITIONAL

- Software & Analysis Skills:** ANSYS Fluent (CFD), SolidWorks, Fusion 360, Python, ANSYS Mechanical (FEA), MATLAB, XFOIL
- Test & Instrumentation Skills:** HITL testing, CAN bus telemetry, power supplies, load banks, multimeters, thermocouples, thermal imaging, micrometers, wind-tunnel testing, pitot tubes, pressure sensors
- Manufacturing Skills:** Additive manufacturing, sheet-metal fabrication, machining, wet layups, resin infusion
- Honors:** National Merit Finalist, AP Capstone Diploma, VHS "Second to None" Award, AP Scholar with Distinction
- Language Fluency:** Conversational Spanish
- Volunteering Organizations:** Texas Iron Spikes, Young Men's Service League, Power for Parkinson's
- Work Eligibility:** US Citizen; eligible for ITAR-controlled work